
FIELD MANUAL

Engulfs

How to play, the eight skills a round is won with, and the technique behind them.

Arena	8,800 x 8,800 units, 4,100 orbs, 90 viruses
Round	10 minutes, top 5 are paid
Field	100 players in Standard, 50 in High stakes
Stake	1.00 or 2.00 USDC, or play free in Practice

Every figure in this manual is read straight from the simulation the server runs. Where a number appears, that is the number in force.

The game in ninety seconds

You are a cell. You eat orbs to grow, you eat players smaller than you, and players larger than you eat you. The round runs for 10 minutes. When the whistle goes, everyone still alive is ranked by mass and the top 5 are paid.

How a round runs

ONE CYCLE

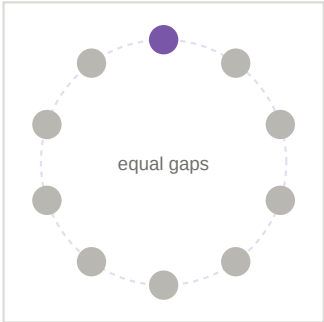
Lobby	Count 5s	Round 10 min	Standings 15s
until the room fills	everyone ready	the only part that pays	opt in for the next

Readiness carries over, so a full lobby rolls straight from the standings into the next count.

You start on a ring

Every player opens on one circle at equal spacing: the same distance to either neighbour, the same distance to the centre, the same distance to the wall. Nobody is handed a corner to themselves and nobody opens inside somebody's mouth. In a full Standard lobby that gap is about 267 units.

The ring is turned by a fresh angle every round, so there is no memorised opening to learn.



Three things end your round

- **You are eaten.** Everything you were carrying goes to whoever ate you.
- **You touch a virus while over 126 mass.** You burst into 9 pieces and become food for everyone nearby.
- **The clock runs out.** The only ending that pays.

The one rule that decides everything else

You can eat another cell only when you are at least **9.8% bigger** than it. Not a little bigger - 9.8%. Against a rival of mass 100 you need 109.8. Everything in this manual is downstream of that number: who you chase, who you run from, and how close you dare sit.

What is in here

Controls	Four inputs, and the HUD	The wager	What is at stake, and when
The rules	Eating, speed, decay	Eight skills	The habits a round is won with
Splitting	What it buys and what it costs	Technique	How the work is actually done
Ejecting	The only thing worth spending on	Playbook	A shape for ten minutes
Viruses	Cover, mine, and weapon	Reference	Every number on one page

Controls

Four inputs. The difficulty is entirely in when you use them.

ACTION	KEYBOARD	TOUCH	WHAT IT DOES
Move	Mouse	Drag	Your cells steer toward the pointer. There is no stop button - you are always moving.
Split	Space	Split	Every cell splits in half and the new halves are launched forward. Up to 16 cells.
Eject	W	Feed	Throws 13 mass out of the front of each cell, at a cost of 16.
Spectate	-	Arrows	Once you are dead, watch anyone still alive.

What the screen is telling you

READOUT	WHY YOU CARE
Clock, top centre	Time left. It turns red for the last 30 seconds, which is the only moment placing matters more than growing.
Rank, top left	Your position and the size of the field. Above 5 is the whole game.
Leaderboard, top right	The top ten, with a line ruled under 5th. That line is the pay boundary - watch who is sitting on it.
At risk / mass value	What you are carrying, and what your mass is nominally worth (200 mass reads as 1.00 USDC). Shown in wagered rounds only.
Minimap, bottom left	Where you are in the arena. Mostly it tells you how close you are to a wall, which is where players get cornered.

What you are actually looking at

Your own cell is predicted locally, so it answers the mouse immediately and does not wait for the server. Everyone else is drawn from what the server last sent, which on a slow connection is a fraction of a second in the past.

The practical consequence: **other players are always slightly further along than they look**. On a contested chase, aim where a target is going rather than where it is drawn. This is also why a fight that felt unfair sometimes was - not because anything is broken, but because you were both acting on slightly different pictures.

Practice first

The Practice tier runs entirely in your own browser against bots. There is no server, no lobby and no money. Every mechanic in this manual behaves identically there, so it is the right place to learn the timing of a split and the feel of feeding a virus - and, because nothing is predicted over a network, the right place to learn what a clean read looks like.

The rules that never change

Three mechanics shape every decision you make. Learn the numbers and most of the game becomes arithmetic you can do at a glance.

1. The 9.8% rule

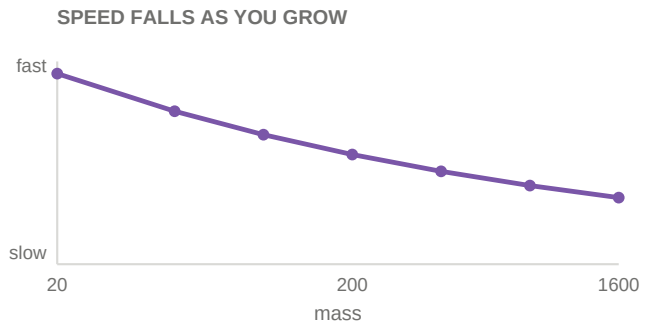


Eating pays a bonus: you collect 1.2x the mass you swallowed, so players are worth far more than orbs. An orb is 3.375 mass, and it takes about 24 of them to go from your opening 20 to 100.

2. Mass costs speed

Speed falls as mass^{-0.24}. Doubling your mass costs you about 15% of your top speed, and it never stops costing. The biggest cell on the board is also the one that cannot catch anything and cannot run away.

This is why the leader is not safe. It is why a mid-sized player can farm the edges of a fight and arrive at the whistle in the places, while the monster that ate everyone gets cornered at minute nine.



MASS	RADIUS	SPEED	VS. START	REJOIN AFTER A SPLIT
20	18	497	100%	20s
100	40	338	68%	21s
200	57	286	58%	23s
400	80	242	49%	27s
800	113	205	41%	35s
1600	160	174	35%	50s

3. Size above a point is a leak

Any cell over **260** mass shrinks continuously, at about 0.22% a second. At 400 that is under a mass a second and barely noticeable. At 1,600 you are losing more than three a second, and standing still is going backwards. Big players have to keep eating purely to stand still.

Splitting

Split and every cell you own halves and fires forward. It is the only way to close distance on something faster than you, and it is the most common way players die.

What it buys

Reach. A launched half travels a few hundred units instantly, which is how you catch a cell that would otherwise outrun you. It also doubles your cell count, up to a ceiling of 16.

What it costs

Everything. Each half is half your mass, so each is eatable by things that could not touch you a second ago. And the pieces cannot rejoin for a cooldown that grows with their size - **20 seconds** at small sizes, **50 at mass 1,600**. You are at your most vulnerable for exactly as long as you are at your biggest.

The question to ask before every split

Not 'can I reach it' but 'what eats the halves'. Look at the board behind your target first. If anything near it is bigger than half of you, the split is a donation.

Ejecting

Eject throws a blob forward. It costs 16 and delivers 13, so about **19% is lost every throw**. You cannot pick your own mass back up for 0.6 seconds, which means recovering it is a decision - turning round and going back - rather than an accident.

There is exactly one thing worth spending that mass on, and it is on the next page.

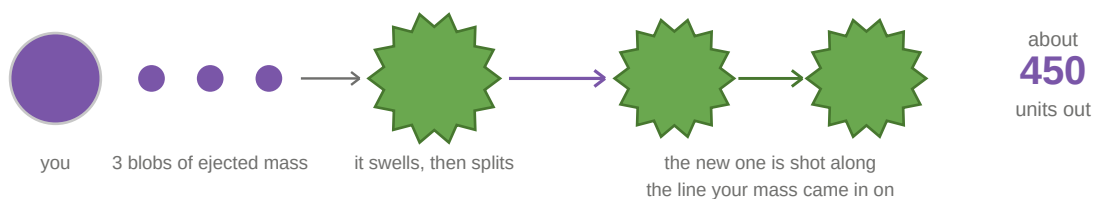
Viruses: the hazard, and the weapon

The green spiked circles. There are 90 of them in a Standard arena, and they behave completely differently depending on how big you are.

IF YOU ARE...	A VIRUS IS...
under 126 mass	Cover. You pass straight over it, and anything big enough to eat you cannot follow you in. Sitting on a virus is the single best defensive move a small cell has.
over 126 mass	A mine. Touch one and you burst into 9 pieces, scattered and unable to rejoin for a long cooldown, in front of everyone who was chasing you. This is how most large players actually die.

Feeding a virus

Shoot ejected mass into a virus and it swells visibly. The **3rd hit splits it**: a new virus is launched along exactly the line your mass came in on, and coasts about 450 units before settling.



This is the only aimed weapon in the game. You are not throwing the virus at someone so much as choosing where you stand while you feed it - the line from you, through the virus, is where the new one goes. Used properly it puts a mine in the path of a player far too big for you to fight, or cuts off the open water a leader was running for.

What it costs you

Three throws is 48 mass off your own total, gone. Feeding is a pure sink: the virus keeps it, you get nothing back. What you buy is position.

Two things that will catch you out

Only mass *still in flight* feeds a virus - a blob that runs out of travel short of it has missed. And if you are over 126 mass, walking close enough to aim comfortably is how you burst yourself. Feed from range, or feed while small.

What is actually at stake

Skip this page if you are playing Practice - nothing below applies. In a wagered round the money is server-side and moves only in response to things that happen in the arena.

WHEN THIS HAPPENS	YOUR MONEY DOES THIS
You join a wagered round	Your stake moves out of your balance and into escrow - your pot. You cannot spend it while the round runs.
You eat a staked player	Their entire pot moves into yours. This is the only way a pot grows.
You are eaten by a staked player	Your whole pot becomes theirs.
You finish in the top 5, alive	Your pot is paid into your balance: your stake plus everything you took off the players you ate.
You finish alive but outside the top 5	Your pot is forfeited. Surviving is necessary; it is not sufficient.
You disconnect and are not eaten	Your body stays in the arena for 6 seconds. Survive that window unclaimed and your stake is refunded.

Death ends your wager, even if you play on

You can respawn and keep playing after being eaten, and you should - position is still worth having. But your pot left with your killer. Finishing in the top 5 after dying once pays out an empty escrow, which is nothing. **Your wager is decided the moment you are eaten, not at the whistle.**

What this means for how you play

- **Mass sets your rank, not your payout.** The figure the HUD shows against your mass is a readout, not a promise. What you are actually paid is your own escrow - stake plus claimed pots.
- **Eating a big staked player is worth more than any amount of farming.** Orbs move you up the leaderboard. Players move your money.
- **A large pot makes you a target.** Late in a round the player who has eaten three staked rivals is carrying four stakes, and everybody can see how big they are.

The eight skills

Rounds are not won by reflexes. They are won by eight separate habits, each of which can be practised on its own. In rough order of how much a beginner gains from fixing it.

1

SURVIVAL

Threat arithmetic

At any moment you should know, without thinking, which cells on your screen can eat you and which you can eat. The line is 9.8%: anything over 1.098 times your mass is lethal, anything under 0.911 times is food.

Beginners read size roughly and get it wrong at the margin, which is exactly where rounds are decided. The margin is narrow on purpose - a cell 5% bigger than you is not a threat, and a cell 10% bigger is.

Drill. In Practice, park next to bots of similar size and read their mass off the leaderboard before you commit. Get used to the visual difference a 10% mass gap makes - it is smaller than you expect, because radius grows with the square root of mass.

2

SURVIVAL

Split discipline

The commonest way to lose a round you were winning. A split halves every cell you own and locks the pieces apart for 20 to 50 seconds.

Good players split when the target is certain and the surrounding board is empty. Bad players split at anything that looks catchable, and spend the next 30 seconds as eight easy meals.

Drill. Play a whole Practice round without splitting once. You will finish smaller but you will learn how much of the board you can hold without it. Then add splits back one at a time, only when nothing bigger than half your mass is within a screen.

3

BOTH

Virus literacy

Know where every virus near you is, and which side of 126 mass you are on. Under it they are your best cover - a big chaser physically cannot follow you onto one. Over it they are the thing most likely to kill you.

The advanced half is offensive: feeding one takes 3 throws and lets you place a new virus roughly 450 units away along a line you choose.

Drill. Grow past the threshold in Practice and deliberately run a bot chase near viruses until you can feel the danger radius. Then spend a round doing nothing but feeding viruses, to learn the range and lead of a throw.

4

GROWTH

Orb economy

Orbs are 3.375 mass each and there are 4,100 of them. Early on, they are the whole game: you need roughly 24 to reach 100 from your opening 20.

The skill is pathing - sweeping through dense clusters in straight lines rather than chasing individual orbs, and doing it in the quiet parts of the arena while the aggressive players trade early deaths.

Drill. Time yourself from spawn to mass 100 in Practice. A good line beats a greedy one; compare a straight sweep against darting at whatever is nearest.

5

BOTH

Speed awareness

Every mass you gain makes you slower, on a curve that never flattens. By mass 400 you are at roughly half your opening speed; by 1,600, a third.

Two consequences people miss. You cannot catch anyone much smaller than you without splitting, so growth past a point converts into pure vulnerability. And when you are the biggest thing on the board, running away has stopped being an option - you have to fight or use viruses.

Drill. Grow to 800 in Practice and try to catch a starting-size bot. You cannot. Learn that before a real round teaches you.

6

SURVIVAL

Map and wall sense

The arena is square and the walls do not push back. A cell pinned against one has lost half its escape directions, and experienced players herd others into corners deliberately.

Keep the middle of the board available to you. The minimap exists for exactly one question: how much room is behind me?

Drill. Watch your minimap dot rather than the arena for thirty seconds in Practice. Notice how often you drift into an edge without deciding to.

7

WINNING

Clock management

The round is 10 minutes and only the last moment of it counts. Rank at minute four is worth nothing; rank at 0:00 is worth everything.

Early round, take risks - a death at minute two costs you a stake you can re-enter with next round. Late round, with a place in hand, every fight you do not need is a fight that can only cost you. The last 30 seconds are for defending a position, not improving it.

Drill. In the final minute, find the line under 5th on the leaderboard and play only against the players either side of it. Everything else on the board is irrelevant to you.

8

WINNING

Pot awareness

Only in wagered rounds, and it is what separates a good player from a profitable one. Your payout is your escrow, and your escrow grows only by eating other staked players.

So a round spent farming orbs to a safe fourth place returns your stake and little else. A round in which you took two staked players and finished fifth pays three stakes. Rank decides whether you are paid; kills decide how much.

Drill. After each wagered round, ask which of your fights changed your escrow and which merely changed your rank. Most players find they spent the round on the second kind.

Technique

The skills chapter is what to work on. This is how the work is actually done - eight techniques, and the numbers each one turns on.

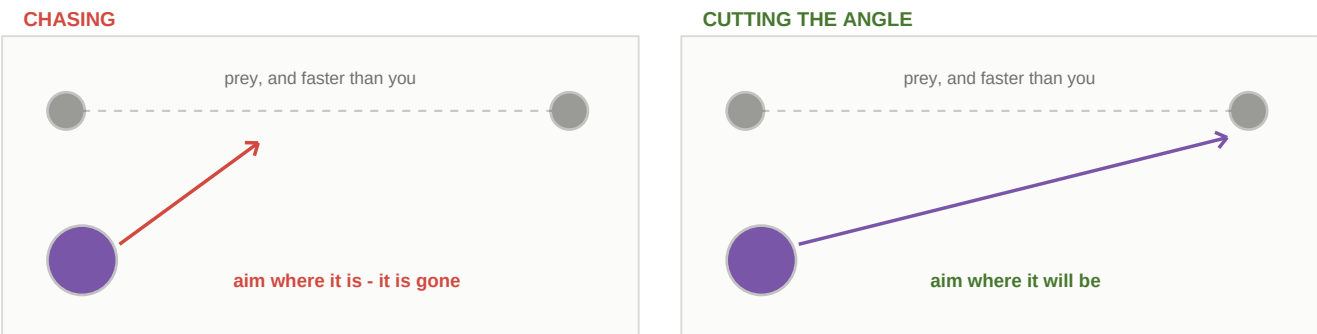
Movement and positioning

Avoiding larger cells while creating angles on smaller players.

Start from the fact that makes every chase futile: **every cell you are big enough to eat is faster than you are.** Speed falls with mass, so being able to eat something guarantees it can outrun you. Even the marginal target - the one only 9.8% smaller - has the legs on you.

YOU ARE MASS 400 AND THEY ARE	THEIR SPEED	YOURS	THEIR ADVANTAGE
364 (the biggest you can eat, just)	248	242	2.3% faster
308	258	242	6.5% faster
200	286	242	18.1% faster
100	338	242	39.5% faster
50	399	242	64.7% faster

So pursuit is not a tactic, it is a way of losing ground slowly. What works is geometry: take the inside line, aim at where the target must go rather than where it is, and push it toward something that limits its options - a wall, a virus, or a bigger player.



Against larger cells the same geometry runs backwards. Keep two escape directions open at all times; the moment you have one, you are being herded. Open water behind you is worth more than orbs in front of you.

The tell you got it wrong. You spent ten seconds behind the same player and never closed the gap. That was never going to work - you needed an angle, a wall, or a split, from the first second.

Split timing and accuracy

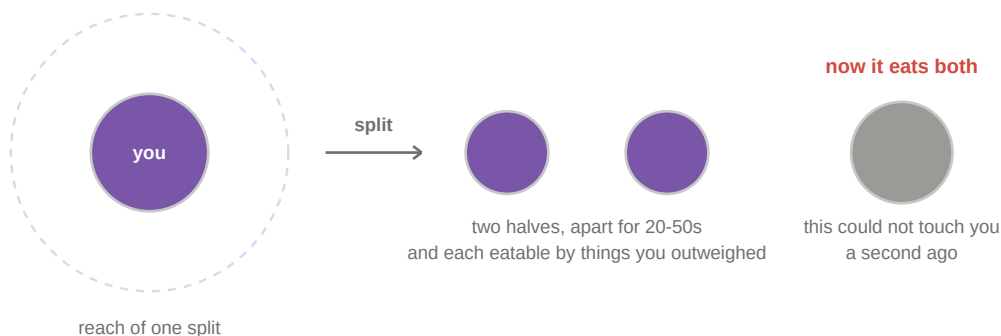
Judging whether a split will reach and consume another player without leaving you vulnerable.

The split is the only thing that closes distance, because the launch is roughly **5.6 times your cruising speed**. It is also a timer on your own vulnerability. Both halves of the decision are in one table.

YOUR MASS	EACH HALF	REACH FROM YOUR CENTRE	YOU STAY IN PIECES FOR
60	30	about 282 units	20s
100	50	about 308 units	21s
200	100	about 355 units	23s
400	200	about 422 units	27s
800	400	about 517 units	35s
1600	800	about 651 units	50s

Three questions, in order

- **Will it reach?** Read the table, not your instinct. At mass 200 you throw about 355 units; at 800, about 517. Anything beyond that is a donation.
- **Will the half be big enough?** The piece that arrives is half your mass, and it still needs to be 1.098x the target. Halving 200 gives you 100, which eats nothing above 91.
- **What eats the halves?** The question people skip. Look behind the target before you commit, not after.



The tell you got it wrong. You landed the split, ate the target, and were eaten yourself before the pieces rejoined. The split worked and the decision did not.

Mass management

Deciding when to split, merge, feed, or preserve mass.

You have four things you can do with mass, and each is a trade you should be able to price.

DO THIS	YOU PAY	YOU GET
Split	Half your size per cell, and 27s apart at mass 400	Reach, and a second mouth
Merge	Nothing, but you cannot hurry it	Back to one fast, safe body
Feed a virus	16 mass a throw, 19% of it lost outright	A hazard placed where you choose
Preserve	Nothing below 260 mass; above it you leak 0.22% a second	Speed, and the ability to run

The non-obvious one is **preserve**. Because speed falls with mass and everything above 260 decays, there is a size past which extra mass makes you worse at everything except eating people who cannot escape. Growing beyond it is a decision, not a reward - take it when you intend to hunt, not because the orbs were there.

Mass in pieces is not mass

Sixteen cells of 50 is not a 800-mass player. It is sixteen 50-mass players who happen to share a name, each of which can be eaten separately, and none of which can rejoin for 35 seconds. Count your effective size as your *largest cell*, not your total.

Risk assessment

Deciding when a target is worth pursuing versus when to retreat.

Price the attempt before you start it. A pursuit costs time, position and usually a split; what it returns is 1.2x the target's mass, plus - in a wagered round - their entire pot.

- **Worth it:** a staked player carrying pots, cornered, with nothing bigger than half your mass within a screen.
- **Not worth it:** anything in open water that can see you coming. It is faster than you and the chase is free for it.
- **Never:** a target sitting on or beside a virus while you are over 126 mass. That is the trap, and good players set it deliberately.

Map awareness

Tracking nearby threats, viruses, escape routes and crowded areas.

Four things to keep a running count of, in this order of urgency: **what can eat me**, **where the viruses are**, **which way is open**, and **where the crowd is**.

The crowd matters more than beginners expect. A fight between two large players generates fragments, ejected mass and burst cells - a burst player scatters into 9 pieces - and the profit is in arriving after it, not during. Equally, a quiet quadrant is where you grow unmolested for the first two minutes.

How far you can see is not a matter of opinion. Everyone is shown the same **760 units** in every direction, and that does not change until you pass about **446 mass** - past which your view finally starts growing with your size.

Two things follow. For most of a round every player on the board has exactly your horizon, so anything you cannot see cannot see you either. And the handful of players large enough to have outgrown the floor are the only ones with an information advantage - which is part of why a leader is hard to ambush.

The tell you got it wrong. You died to something you never saw. Nine times in ten that is a wall at your back, not bad luck.

Prediction

Anticipating opponents' movement and split attacks.

Two things to read.

Movement. Everyone else is drawn from the last thing the server sent, so on a slow connection you are seeing them a fraction of a second in the past. Lead your targets. This cuts the other way too: your own cell is predicted locally and is exactly where you think it is, which is why your escapes work better than your chases.

Split attacks. A player about to split lines up first - they stop cutting corners and start travelling straight at you. The warning is short and the reach is long (422 units at mass 400), so the counter is distance held in advance, not reaction. If something twice your size is pointing at you and closing, you are already inside its range.

The counter nobody uses

A player who has just split is two half-sized cells that cannot rejoin for tens of seconds. If they miss you, they are prey - and for that window they are the most profitable target on the board. Baiting a split and turning on the pieces is the highest-value play in the game.

Virus mechanics

Skilled players use viruses offensively and defensively.

Defensively, while you are under 126 mass, a virus is a place nothing dangerous can follow you. Learn where they are and treat them as safe squares - a chase that would kill you in open water ends the moment you sit on one.

Offensively, there are two plays. The first needs no setup: herd a large player toward a virus. They know the danger, so the threat of it alone steers them, which is often all you want.

The second is feeding. 3 blobs of ejected mass split a virus and launch a new one along the line your mass came in on, about 450 units. That is how a small player removes a large one without ever being big enough to fight it - and how you close off the open water someone was running for.

- Cost: 48 of your own mass, unrecoverable.
- Only mass still in flight counts, so feed from a distance you can actually throw - a blob that stops short has missed.
- Do not set up a feed while over 126 mass without watching your own drift. Walking in to aim is how people burst themselves.

Team tactics

Feeding, baiting, splitting and coordinating - where applicable.

The mechanics exist, so it is worth knowing what they are. **Feeding** transfers mass to another player at 19% loss per throw. **Baiting** means one player presenting as catchable so a third party splits at them, and the partner taking the pieces. **Coordinated splitting** means two players cutting off both escape directions at once, which defeats the geometry in the positioning section above.

Where this applies, and where it does not

In Practice, and in any friendly round, all of the above is fair play and good fun. **In a wagered round it is not a tactic, it is collusion.** Two accounts moving stakes between themselves by feeding is precisely the behaviour a staked game has to exclude, and the fact that the mechanics allow it is an open problem rather than a licence. Treat wagered rounds as what they are: every other player is an opponent.

The one piece of it that is always legitimate is reading other people's coordination. If two cells keep arriving together, assume they are working together and stop treating either as a lone target.

A ten-minute playbook

One way to structure a round. Not the only way, but a shape to depart from rather than playing on instinct for ten minutes.

CLOCK	PRIORITY	WHAT YOU ARE DOING
10:00 - 8:00	Grow, quietly	You open equidistant from two neighbours who are exactly your size, so nobody can eat anybody. Leave. Sweep orbs away from the ring and let the players who fight at the opening whistle eat each other.
8:00 - 5:00	Convert size into kills	You should be comfortably over 100. Hunt players in the band you can actually eat - anything under 0.91x your mass - and take them without splitting where possible. In a waged round this is the only part that grows your pot.
5:00 - 2:00	Manage the crown	If you are large, you are slow and you are decaying above 260. Stay off viruses, stay off walls, and keep eating enough to cover the leak. If you are mid-sized, pick off the fragments from other people's fights - a burst player is nine free meals.
2:00 - 0:30	Find the line	Look at where 5th place sits. If you are above it, your job is no longer to grow. If you are below it, you need one player, and viruses are how a smaller cell removes a bigger one.
0:30 - 0:00	Survive	Do not split. Do not chase. Do not go near a virus. Hold open water and let the clock run out.

The five mistakes that cost the most

- **Splitting to catch something you did not need.** The halves are what gets eaten, and the cooldown is measured in tens of seconds.
- **Carrying more than 260 mass without a plan.** Above it you leak, you are slow, and viruses are lethal to you.
- **Chasing a rank instead of a pot.** Rank without kills pays back a stake and a shrug.
- **Fighting in the last thirty seconds.** A place in hand is worth more than a place you might win.
- **Treating a respawn as a second chance at the money.** It is not. Your pot left with the player who ate you.

Quick reference

Every number the game runs on, on one page.

Eating and growing

Starting mass	20
Orb value	3.375 mass
To eat a rival	you must be 1.098x their mass (9.8% bigger)
Reward for eating	1.2x the mass you swallowed
Decay	above 260 mass, 0.22% per second
Speed	falls as $\text{mass}^{-0.24}$

Splitting and ejecting

Maximum cells	16
Rejoin cooldown	about 20s at small sizes, 50s at mass 1,600
Eject cost	16 mass spent, 13 delivered (19% lost)
Eject range	about 237 units past your own membrane
Own-mass cooldown	0.6s before eating your own throw back

Viruses

Virus mass	110
Dangerous to you above	126 mass
Pieces you burst into	9
Feeds to split one	3 blobs of ejected mass, still in flight
Where the new one goes	along the line your mass came in on, about 450 units
Population ceiling	1.5x the count the arena seeds

The round

	STD	HIGH
Stake	1.00	2.00
Players needed	100	50
Arena	8,800	6,200
Orbs	4,100	2,035
Viruses	90	45
Round length	10:00	10:00
Paid places	5	5
Countdown	5s	5s
Standings	15s	15s

Practice is a third tier with no stake, no lobby and no server: it runs against bots in your own browser. Use it for everything in the skills chapter.